Sahaj Bhadja

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Summary

Software engineer with 2+ years building distributed services, asynchronous pipelines, and data-intensive applications using Python, Java, JavaScript/TypeScript, SQL, algorithms, APIs, performance optimization, testing, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, OOP, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant Azure SQL and Blob-backed services for 500+ users, optimizing indexed queries, schemas, and access paths to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant asynchronous workflows for 1,000+ monthly jobs using explicit states, retries, validation, and failure isolation, improving throughput 3x while strengthening monitoring, debugging, and recurring production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Flask REST APIs and React workflows for secure ingestion and automated analysis, supporting 10,000+ daily requests through clean interfaces, reusable components, documented services, and reliable full-stack behavior.
- Implemented multithreaded processing and MySQL on AWS RDS for 1,000+ uploads per batch, reducing turnaround by 60% while improving testability, stability, and troubleshooting across concurrent production workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python pipelines processing 100,000+ autonomous-driving trajectory logs, implementing validation, feature preparation, and repeatable experiments that improved long-horizon planning performance by 18% across large-scale simulation benchmarks.
- Developed spatial-temporal encoders combining map geometry, behavioral context, and multi-head attention, improving closed-loop stability by 15% while analyzing architectural trade-offs and debugging complex production data-dependent model failures.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform microservices and 20+ automated unit and integration suites using Jenkins, PyTest, and Mockito, improving code-review confidence, release reliability, and debugging speed across Windows, Linux, and macOS.
- Optimized REST API and ETL services with OOP patterns, AWS Glue, S3, and SQL processing, improving response times by 30% and pipeline efficiency by 68% for production data operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN platform with REST APIs, WebSockets, and modular Node.js services, supporting 2M+ synchronized records while improving database access, real-time updates, and full-stack coordination across distributed users.
- Automated Docker-based CI/CD testing and deployments with Jenkins, cutting release time by 50% while improving build consistency, regression coverage, and troubleshooting for frequent service changes and production rollouts.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, Bash; **Backend**: Flask, FastAPI, Node.js, Express.js, REST APIs, WebSockets, microservices, OOP; **Frontend**: React.js, Next.js, HTML5, CSS3
Data & Systems: PostgreSQL, MySQL, Azure SQL, MongoDB, AWS RDS, indexing, distributed systems, asynchronous processing, TCP/IP; **Engineering**: data structures, algorithms, Git, Docker, Jenkins, CI/CD, unit testing, integration testing, code reviews, monitoring, performance optimization, debugging, Agile/Scrum

Projects

Big-Brain Live | *Python, PostgreSQL, REST APIs, FAISS, LLaMA-2*
- Built a semantic retrieval platform over 50,000+ documents using transformer embeddings and FAISS, enabling contextual search and summarization with 95% query accuracy through optimized indexing and similarity pipelines.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes*
- Built a scalable platform for 50,000+ users and 1M+ transactions, combining secure APIs, indexed data access, real-time updates, and Kubernetes-backed deployments for reliable high-volume customer operations.

CPU-Scheduling Simulator Live | *C++, React, JavaScript*
- Implemented FCFS, SJF, SRT, RR, priority scheduling, and multilevel queues with dynamic metrics and Gantt visualization, comparing algorithmic performance across 1,000+ simulated processes under varied workloads.